

Water 1
Infrastructure Discipline
TONKIN & TAYLOR

Variational Autoencoder for Climate Change-Driven Hydrological Stochastic Data Generation

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Acknowledgements

I would like to acknowledge ...

Declaration

I declare that this report and all in it has been written independantly by myself.

Martin Wright, October 2025

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Chapter 1

Introduction

Global Climate Models (GCMs) provide the primary basis for understanding future climate under alternative emissions scenarios, yet their coarse spatial resolution limits direct application for regional impact assessments. Downscaling techniques bridge this gap by translating GCM outputs to the finer scales required by local decision-making processes. For Auckland’s metropolitan water supply system, this is critical for long-term planning.

Watercare operates the Integrated Source Management Model (ISMM) to optimise conjunctive use of their water reservoirs, groundwater sources, and rivers. The ISMM program requires long daily sequences of rainfall and potential evapotranspiration (PET) to evaluate system yields under current and future climate conditions.

Previous work for Watercare developed a methodology combining dynamically downscaled GCM projections with stochastic weather generation. Historical rainfall records were perturbed using shifts in gamma distribution parameters derived from downscaled projections, then extended using the Stochastic Climate Library (SCL) to produce 1000-year datasets. This approach has enabled assessment of water supply resilience under Representative Concentration Pathways 4.5 and 8.5 for mid-century (2040) and late-century (2090) time horizons, providing valuable insights for infrastructure planning.

However, several limitations constrain the robustness of this framework for future assessments. Perturbation methods based on fitted parametric distributions struggle to represent extreme rainfall events that are rare in historical records but critical for water supply security. Stochastic weather generators reproduce statistical variability from training data rather than generating physically plausible extremes beyond observed ranges. Traditional generators also treat daily weather as conditionally independent given the previous day, failing to capture the full temporal coherence of multi-day storm systems and drought evolution. Furthermore, the existing framework incorporates PET as simplified annually repeating functions, weakening representation of compound climate stressors such as concurrent drought and high evaporative demand.

Recent advances in deep generative models offer alternative approaches that may address these limitations. Variational Autoencoders (VAEs) learn flexible representations of complex data distributions without assuming parametric forms, and their structured latent spaces enable controlled generation of samples with varying characteristics. This research extends the VAE framework by incorporating learned latent dynamics to capture temporal evolution of weather patterns. Unlike standard VAEs that treat each observation independently, sequential VAEs learn how latent representations evolve from one day to the next, enabling generation of temporally coherent multi-day sequences.

The model is trained on bias-corrected, dynamically downscaled GCM projections for both historical and future scenarios, learning spatial weather patterns, their temporal evolution, and how these characteristics shift under climate change. By combining strategic latent space sampling with learned dynamics and temporal conditioning, the approach aims to generate thousand-year synthetic datasets suitable for ISMM while addressing key limitations of existing methods. The research develops a sequential VAE architecture for multi-variable spatial weather fields, implements recurrent latent dynamics using Gated Recurrent Units, and demonstrates controlled generation of extreme scenarios through latent space sampling. The ultimate objective is to provide Watercare with enhanced capability for assessing water supply resilience under climate change, particularly for

extreme events that pose greatest risk to system reliability yet remain poorly represented in traditional synthetic weather datasets.

Chapter 2

Background

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2.1 Auckland’s Water Supply System and ISMM

Auckland’s metropolitan water supply serves over 1.7 million people across New Zealand’s largest urban area. The system comprises multiple sources managed conjunctively: surface water reservoirs in the Hunua and Waitakere Ranges, groundwater from various aquifer systems, and river abstractions from the Waikato River. This diversity provides resilience against individual source failures but requires sophisticated operational management to balance competing demands, environmental constraints, and storage dynamics across the integrated system.

The Integrated Source Management Model (ISMM) is Watercare’s primary decision-support tool for long-term planning and operational optimisation. The model simulates daily water movements through the entire supply system, tracking reservoir storage levels, aquifer drawdown, river flows, and treatment plant capacities. Environmental flow requirements, infrastructure constraints, and operational rules govern how water can be extracted and transferred between sources. To evaluate system performance—particularly the reliable yield, defined as the maximum demand that can be met with acceptable security of supply—the ISMM requires input sequences spanning centuries. This length is necessary to adequately sample low-frequency variability in rainfall and to estimate the probability of severe multi-year droughts that could challenge system resilience.

Previous climate change impact assessments for Watercare relied on producing these extended sequences through a two-stage process. Dynamically downscaled GCM projections provided daily rainfall and temperature data at approximately 5 km resolution across the supply catchments. These projections were bias-corrected against observed records to improve agreement with historical statistics. Monthly gamma distributions were then fitted to both historical and projected rainfall series, and the shifts in distribution parameters between time periods were applied to perturb the observed historical record. This perturbation preserved the day-to-day temporal structure of observed weather while embedding projected changes in rainfall statistics. The perturbed records were then extended using the Stochastic Climate Library, which generated multiple stochastic replicates that were concatenated to produce sequences exceeding one thousand years. This methodology provided workable datasets for ISMM and yielded insights into system vulnerability under RCP4.5 and RCP8.5 scenarios.

2.2 Limitations of Current Methodology

The perturbation-based approach assumes that future rainfall can be adequately represented by shifting the parameters of distributions fitted to historical data. For bulk statistics such as monthly means and standard deviations, this assumption is reasonable and well-supported by theory. However, extreme events occupy the tails of rainfall distributions where sample sizes are inherently small. Fitting parametric distributions to limited extreme samples and then extrapolating through parameter shifts introduces substantial uncertainty. The gamma distribution, while flexible, may not accurately represent the tail

behaviour of precipitation, particularly for sub-daily or multi-day accumulations where extreme value theory suggests alternative distributions may be more appropriate.

Further, one of the arguments for the augmented delta-shift method was that GCMs couldn't model "unknown unknowns" [?]. However, this is not actually true; for example, volcanic forcings (one of the examples explicitly given) *are* actually modelled during the historical calibration run.

The SCL faces a practical constraint regarding multivariate generation: it can either maintain spatial co-variance across sites for a single variable, or temporal co-variance between multiple variables at a site, but not both simultaneously within its current framework. For the Watercare application, accurate spatial co-variance of rainfall across the supply catchments was deemed more critical than dynamic coupling between rainfall and PET. Consequently, PET was incorporated as monthly averaged values that repeat annually, scaled according to future projections but lacking day-to-day variability. This simplification weakens the ability to model compound events where anomalously high temperatures and evaporative demand coincide with rainfall deficits, amplifying drought stress. Such compound events are particularly concerning under climate change, where rising temperatures are projected to increase atmospheric water demand even as precipitation patterns shift.

2.3 Challenges with Bias Correction

Bias correction has become standard practice in climate impact modelling, with quantile mapping (QM) and its variants widely adopted across sectors. These methods align GCM output distributions with observed distributions by mapping quantiles from the modelled data, to quantiles in the observed data. When applied to future projections, the method assumes that the biases present in historical simulations persist into the future, allowing the same transfer functions to correct future data. This assumption of bias stationarity is convenient but physically questionable, as the sources of GCM biases, such as insufficient resolution of orographic precipitation or imperfect representation of convective processes may change as the climate state changes.

Recent papers such as Chandel et al. [?] challenge this approach. They demonstrated that aggressive bias correction can distort or suppress the climate change signals that GCMs are designed to project. Their analysis showed that quantile-mapped GCM fields can become statistically indistinguishable from quantile-mapped random fields, suggesting that the correction process forces GCM outputs to conform so closely to historical observations that future climate signals are obscured. The climate change signal in a GCM projection emerges from the model's representation of radiative forcing, circulation changes, and thermodynamic responses. When quantile mapping forces future distributions to align with historically observed distributions, it may inadvertently remove physically meaningful changes in variability, extreme event frequency, or relationships between variables.

This tension between improving historical fidelity and preserving future climate signals represents a fundamental challenge for impact modelling. Uncorrected GCM outputs often contain substantial biases that render them unsuitable for direct use in impact models calibrated to historical conditions. Yet aggressive correction risks undermining the very climate change information the GCMs provide. The optimal approach likely lies in methods that improve GCM outputs without completely subordinating them to historical statistics, preserving the physical relationships and change signals while reducing systematic biases.

Chapter 3

Literature Review

3.1 Machine Learning for Weather and Climate

The application of machine learning to weather and climate problems has expanded rapidly, driven by increasing computational power, availability of large datasets, and algorithmic innovations. Early applications focused on pattern recognition and classification tasks such as identifying weather regimes or detecting extreme events in climate model output. More recently, generative models have been applied to weather prediction, downscaling, and synthetic data generation.

Variational Autoencoders have emerged as a powerful framework for learning complex probability distributions and generating samples from high-dimensional data. VAEs explicitly model data distributions through a probabilistic encoder-decoder structure, providing both training stability and a structured latent space that enables controlled generation. The regularization of latent space to a known distribution creates natural mechanisms for sampling and interpolation. Oliveira et al. [?] demonstrated VAE application to precipitation field generation, showing that latent space sampling could control scenario extremity. Their approach trained a VAE on monsoon precipitation data, then generated extreme scenarios by sampling from the distribution tails. Quantile-quantile plots validated that samples from low-probability latent regions produced precipitation fields with characteristics matching observed extreme events. This established the principle that VAE latent space structure enables controlled generation of extremes without requiring explicit conditioning variables or separate extreme value models.

The standard VAE framework treats each observation independently, which is suitable for generating spatial fields but cannot capture temporal dependencies essential for multi-day weather sequences. This limitation motivates extension to sequential VAEs that incorporate learned temporal dynamics, enabling generation of coherent time series rather than independent snapshots. For climate applications, machine learning approaches offer both opportunities and challenges. The ability to learn complex non-linear relationships directly from data without restrictive parametric assumptions is attractive, particularly for representing multivariate dependencies and spatial patterns. However, machine learning models generally require large training datasets, may struggle to extrapolate beyond training data ranges, and provide less transparent representations than traditional statistical models. Validation and interpretation require careful attention to ensure generated data possess appropriate physical constraints and statistical properties.

3.2 Temporal Dynamics

Modelling temporal sequences requires capturing dependencies across time. Recurrent neural networks provide a natural framework for sequential modelling through hidden states that carry information forward through time. The hidden state at each timestep is updated based on the current input and previous hidden state, enabling the network to maintain memory of past observations. Gated Recurrent Units (GRU) address the vanishing gradient problem that limits standard RNNs' ability to capture long-range dependencies. The GRU architecture uses reset and update gates to control information flow, allowing the network to learn which aspects of past states to retain and which to discard [?]. This gating mechanism proves particularly effective for weather sequences

where some features (like large-scale circulation patterns) evolve slowly while others (like local precipitation) change rapidly.

In the context of sequential VAEs, GRU networks model latent dynamics through a transition function that predicts the distribution of latent codes at time t given the latent code at time $t - 1$ and the recurrent hidden state. This architecture captures temporal dependencies at two levels: the latent codes themselves evolve over time, and the hidden state maintains additional memory of the sequence history. The dual representation allows the model to capture both fast timescale dynamics (day-to-day changes in weather) and slower processes (gradual evolution of seasonal conditions or multi-week circulation patterns).

Training sequential generative models faces the challenge of exposure bias. During training with teacher forcing, the model receives ground-truth observations at each timestep, allowing it to correct for any errors made at previous steps. During generation, the model must rely on its own predictions, and errors can accumulate over long sequences. Various approaches address this problem, including scheduled sampling (gradually reducing teacher forcing during training), professor forcing (using adversarial training to match training and generation conditions), and rollout-based objectives that explicitly train on self-generated sequences.

For weather sequence generation, capturing appropriate timescales of variability is critical. Daily weather exhibits persistence due to the movement of synoptic systems with characteristic timescales of days. Seasonal cycles operate at annual timescales. Inter-annual variability from climate modes like the El Niño-Southern Oscillation introduces dependencies spanning years. A successful sequential model must balance these multiple timescales, maintaining short-term coherence while allowing longer-term variability. The latent dynamics architecture provides flexibility to learn these multi-scale dependencies from data rather than imposing them through model structure.

3.3 Generative AI

The term “generative AI” has become widely popularised as a way to refer to state-of-the-art foundational models such as stable diffusion or SoRA. However, the term itself has a meaning more grounded in the idea of a model being able to generate reasonable outputs without the need for complex input. This differs greatly to traditional applications of neural networks prior, where a network of perceptrons learn the optimum weights & biases to transform a complex, high-dimensional input into an output. Put simply, $f : x \rightarrow y$.

3.3.1 Variational Autoencoders

The traditional autoencoder is made up of the encoder and decoder, where the decoder output shape matches that of the input into the encoder. The encoder takes the input x and ‘reduces’ it down into some value z , which represents some point in a d dimensional abstract space referred to as “latent” space.

Much noise has been made lately about researching the application of generative adversarial networks (GANs) and stable diffusion models for gridded weather synthesis, due to their successes in the image generation space. The issue with this thinking, however, is that weather synthesis is done on a much coarser grid and for discretely different purposes. While the outputs may experience fuzzing, this smoothness is representative of the inherent variance in weather patterns, and smooths out over long simulations. Perfect for simulating climate change impacts over long time frames.

The variational autoencoder utilises stochastics to improve the performance of an autoencoder by enriching this latent space into a prior and posterior distribution, forcing it to its Gaussian representation [?].

β -VAE

While the VAE achieves limited disentangling performance, it does not scale to more complex datasets such as image generation. The β -VAE is a deep unsupervised approach which augments the standard VAE with the β hyperparameter that modulates the learning constraints being applied to the model [?]. The case $\beta = 1$ matches the performance of the VAE, but for cases $\beta > 1$, the model learns a more efficient latent representation of the input data.

Chapter 4

Scope

Chapter 5

Methods

This study develops a hybrid Variational Autoencoder–Generative Adversarial Network (VAE-GAN) framework for generating synthetic rainfall and potential evapotranspiration (PET) sequences for use in Watercare’s Integrated Source Management Model (ISMM). The framework integrates information from two sources: (i) historical observations and re-analysis datasets, which provide the empirical basis for learning spatial–temporal weather patterns, and (ii) climate projections from downscaled Global Climate Models (GCMs), which provide the climate change signals that must be preserved in the synthetic sequences.

The methodology consists of three stages. First, a VAE is trained using historical fields to learn a structured probabilistic latent space representation of rainfall and PET. Second, GCM-derived perturbations are mapped into the latent space through bias correction and delta-shift methods. Third, a VAE-GAN decoder is employed to generate synthetic weather fields that reflect both the historical climatology and projected climate signals, while the adversarial discriminator improves realism and fidelity, particularly at the extremes.

5.1 Data

The downscaled GCM data was provided by the ministry for the environment [?] [?]. From the CMIP 6 ensemble, six models were chosen and dynamically downscaled. These models were chosen based on their performance over the historical baseline and assessed using specific, technical performance. Here, Gibson et al. [?] look at process-based metrics, model independence and spread in equilibrium climate sensitivity.

5.1.1 Dynamic Downscaling

The models were downscaled with the Conformal Cubic Atmospheric Model (CCAM) [?]. This method uses a "variable-resolution conformal-cubic grid" to recreate a finer resolution over the region of interest alongside a, relatively, high resolution (12-35km) grid over the wider Pacific area.

The first three of the chosen six models (ACCESS-CM2, EC-Earth3, NorESM2-MM) used CCAM through spectral nudging [?]; in other words, the regional climate model (RCM) is biased towards the GCM data by only "nudging" the long wavelengths/large-scale features of the RCM’s simulated fields to those of the GCM. Here, the RCM is biased towards the atmospheric fields, sea surface temperatures and sea ice concentrations of the GCM.

The three other models (AWI-CM-1-1-MR, CNRM-CM6-1, GFDL-ESM4) were simulated using free- running atmospheric calculations but with sea surface temperature and sea ice concentrations being provided externally by the larger GCM.

5.1.2 Downscaled GCM’s

The ‘*National climate projections for Aotearoa New Zealand*’ project produced dynamically downscaled climate projections, as described above in subsection ??.

5.2 Model Architecture

Oliveira et al. [?] included temporal modelling directly into the VAE model. What we are proposing here, is instead to use GRU to model the temporal dependencies, $p(z_t|z_{t-1})$.

Chapter 6

Results

6.1 Statistics

From the distribution comparison and quantile-quantile (Q-Q) plot in figure ??

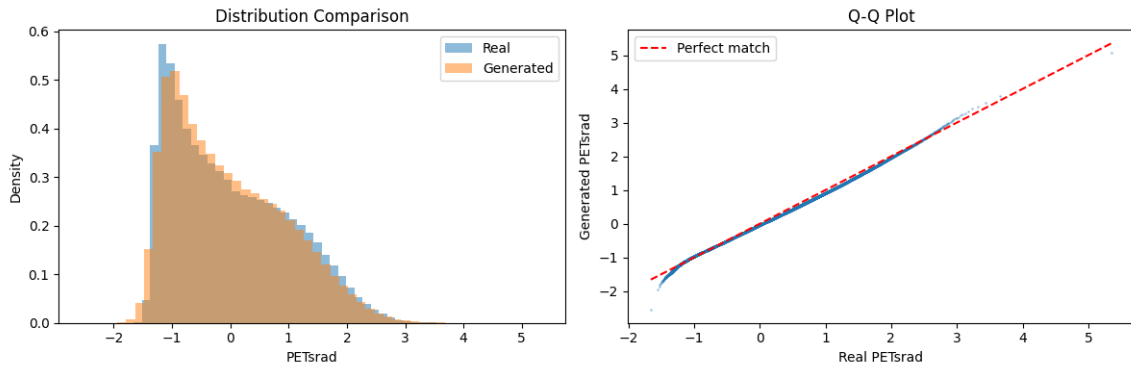


Figure 6.1: Statistical comparison of model training & output data

6.2 Model Outputs

Taking random samples from the historical database and recreating their latent space representations, the following results are produced.

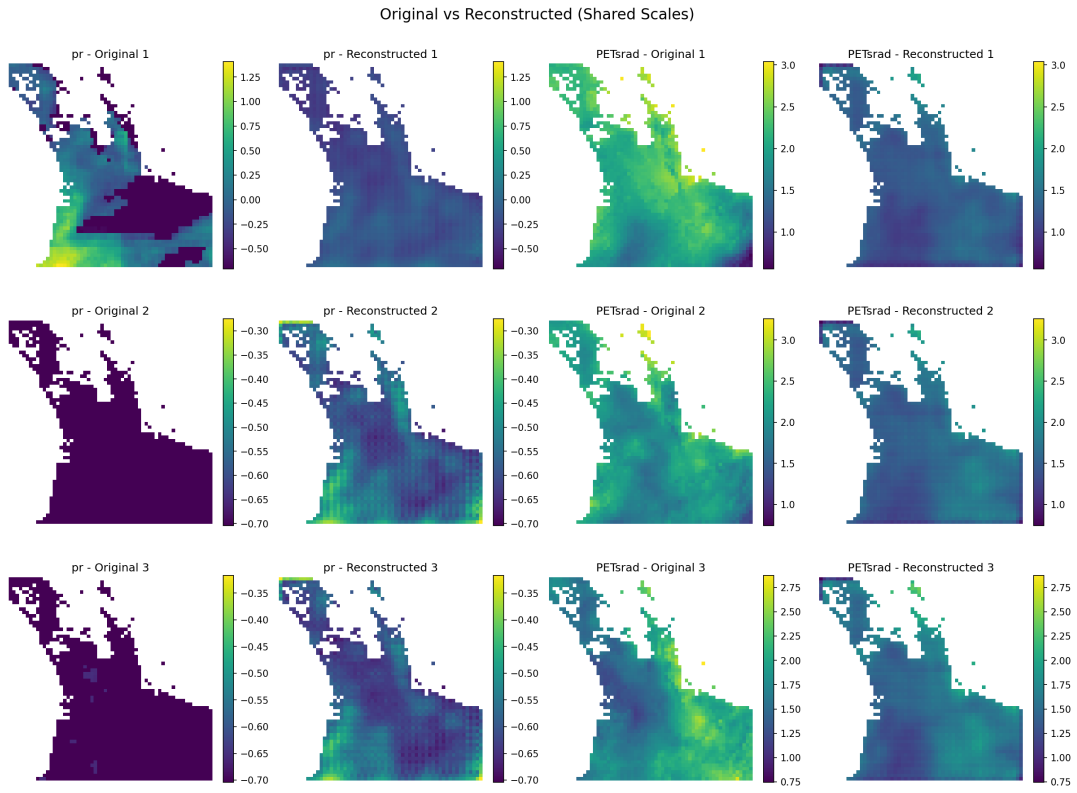


Figure 6.2: VAE Reconstructions of Randomly Sampled Events

Chapter 7

Discussion

Chapter 8

Conclusion

Chapter 9

Appendix A